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Electronic device and method for wireless communication, and computer-readable storage medium

Abstract

Provided are an electronic device and method for wireless communication, and a computer-readable storage medium. The electronic device for wireless communication comprises a processing circuit, which is configured to at least select, from a candidate resource set formed of predetermined candidate time-frequency resource blocks, a time-frequency resource block for the initial transmission and/or retransmission of a data block.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application is based on PCT filing PCT/CN2021/114695, filed Aug. 26, 2021, which claims priority to Chinese Patent Application No. 202010910008.7, filed Sep. 2, 2020, the entire contents of each are incorporated herein by reference in its entirety.
FIELD
(2) The present disclosure relates to the technical field of wireless communications, in particular to a selection of a time-frequency resource block for initially transmitting a data block and/or a time-frequency resource block for retransmitting a data block. More specifically, the present disclosure relates to an electronic apparatus and a method for wireless communications and a computer-readable storage medium.
BACKGROUND
(3) In the existing communication methods, how to select a time-frequency resource block for initially transmitting a data block and/or a time-frequency resource block for retransmitting a data block to reduce communication delay and/or improve communication reliability is a key problem.
SUMMARY
(4) A brief summary of the present disclosure is given hererinafter, to provide a basic understanding to some aspects of the present disclosure. It should be understood that this summary is not an exhaustive summary of the present disclosure. It is not intend to define a key part or an important part of the present disclosure, or to limit the scope of the present disclosure. The purpose is only to give some concepts in a simplified form, which serves as a preface of a more detailed description discussed later.
(5) According to one aspect of the present disclosure, an electronic apparatus for wireless communications is provided. The electronic apparatus includes a processing circuit configured to select a time-frequency resource block for initially transmitting a data block and/or a time-frequency resource block for retransmitting a data block from at least a candidate resource set comprising predetermined candidate time-frequency resource blocks.
(6) According to another aspect of the present disclosure, a method for wireless communications is provided. The method includes selecting a time-frequency resource block for initially transmitting a data block and/or a time-frequency resource block for retransmitting a data block from at least a candidate resource set comprising predetermined candidate time-frequency resource blocks.
(7) According to other aspects of the present disclosure, a computer program code and a computer program product for implementing the above method for wireless communications, and a computer-readable storage medium recording the computer program code for implementing the above method for wireless communications are further provided.
(8) These and other advantages of the present disclosure will be more apparent from the following

detailed description of preferred embodiments of the present disclosure in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to further set forth the above and other advantages and features of the present disclosure, the detailed description of embodiments of the present disclosure will be further described in detail below in conjunction with the accompanying drawings. The accompanying drawings together with the following detailed description are incorporated into and form a part of this specification. Elements with the same function and structure are denoted by the same reference numerals. It should be understood that, these accompanying drawings only illustrate, by way of example, typical embodiments of the present disclosure, and should not be regarded as a limitation to the scope of the present disclosure. In the accompanying drawings:

(2) FIG. **1** shows a block diagram of functional modules of an electronic apparatus for wireless communications according to an embodiment of the present disclosure;

(3) FIG. **2** is a schematic diagram showing a selection of time-frequency resources performed by transmitter user equipment in a sidelink resource selection mode 2 in the conventional technology;

(4) FIG. **3** is a diagram showing an example in which a resource selection window is divided into a predetermined number of sub-resource selection windows in a time dimension according to an embodiment of the present disclosure;

(5) FIG. **4** is a diagram showing an example in which a resource selection window is divided based on the number of candidate time-frequency resource blocks according to an embodiment of the present disclosure;

(6) FIG. **5** is a diagram showing an example in which a resource selection window further includes one or more candidate time-frequency resource blocks included in an abnormal resource pool according to an embodiment of the present disclosure;

(7) FIG. **6** is a diagram showing another example in which a resource selection window further includes one or more candidate time-frequency resource blocks included in an abnormal resource pool according to an embodiment of the present disclosure;

(8) FIG. **7** shows a flowchart of a method for wireless communications according to an embodiment of the present disclosure;

(9) FIG. **8** is a block diagram showing a first example of a schematic configuration of an eNB or gNB to which the technology according to the present disclosure may be applied;

(10) FIG. **9** is a block diagram showing a second example of a schematic configuration of an eNB or gNB to which the technology according to the present disclosure may be applied;

(11) FIG. **10** is a block diagram showing an example of a schematic configuration of a smart phone to which the technology according to the present disclosure may be applied;

(12) FIG. **11** is a block diagram showing an example of a schematic configuration of a vehicle navigation device to which the technology according to the present disclosure may be applied; and

(13) FIG. **12** is a block diagram showing an exemplary structure of a general-purpose personal computer capable of implementing the method and/or apparatuse and/or system according to the embodiments of the present disclosure.

DETAILED DESCRIPTION OF EMBODIMENTS

(14) Hereinafter, exemplary embodiments of the present disclosure will be described in conjunction with the accompanying drawings. For the sake of clarity and conciseness, not all features of the practical embodiments are described in this specification. However, it should be understood that multiple decisions specific to the embodiments have to be made in a process of developing any such practical embodiments, so as to achieve a specific object of a developer; for example,

conforming to those constraints related to a system and a service, and these constraints may change as the embodiments differ. In addition, it should also be appreciated that, although the development work may be very complicated and time-consuming, for those skilled in the art benefiting from the present disclosure, such development work is only a routine task.
(15) Here, it should further be noted that, in order to avoid obscuring the present disclosure due to unnecessary details, only an apparatus structure and/or processing steps closely related to the solution according to the present disclosure are shown in the accompanying drawings, and other details having little relevance to the present disclosure are omitted.
(16) FIG. **1** shows a block diagram of functional modules of an electronic apparatus **100** for wireless communications according to an embodiment of the present disclosure. As shown in FIG. **1**, the electronic apparatus **100** includes a processing unit **101**. The processing unit **101** is configured to select a time-frequency resource block for initially transmitting a data block and/or a time-frequency resource block for retransmitting a data block from at least a candidate resource set comprising predetermined candidate time-frequency resource blocks.
(17) The processing unit **101** may be implemented by one or more processing circuitry, and the processing circuitry may be implemented as, for example, a chip.
(18) The electronic apparatus **100**, for example, may be arranged on user equipment (UE) side or may be communicatively connected to user equipment. Here, it should also be noted that, the electronic apparatus **100** may be implemented at a chip level or at a device level. For example, the electronic apparatus **100** may function as user equipment, and may also include an external device such as a memory and a transceiver (not shown in the figure). The memory may be configured to store programs required for performing various functions by the user equipment and related data information. The transceiver may include one or more communication interfaces to support communication with different devices (for example, a base station, and other user equipment). Implementation of the transceiver is not specifically limited here. The base station may be an eNB or gNB, for example.
(19) As an example, the processing unit **101** may be configured to select a time-frequency resource block for initially transmitting a data block and/or a time-frequency resource block for retransmitting a data block in a scenario of sidelink resource selection mode 2.
(20) In 5G NR sidelink communication, there are two resource selection modes. In one resource selection mode, a base station schedules sidelink resources, which is referred to as resource selection mode 1 (mode 1). In another resource selection mode, UE autonomously selects resources, which is referred to as resource selection mode 2 (mode 2).
(21) FIG. **2** is a schematic diagram showing a selection of time-frequency resources performed by a transmitter user equipment in a sidelink resource selection mode 2 in the conventional technology. In the following description with reference to FIG. **2**, a time-frequency resource is sometimes referred to as a resource.
(22) The transmitter user equipment (TX UE for short) first predetermines a candidate resource set by a resource awareness process. As shown in FIG. **2**, during a resource selection process in the sidelink resource selection mode 2, if a data packet triggers the resource selection at a time instant n, the TX UE excludes resources by using a result awared during the awareness window [n-T.sub.0,n-T.sub.proc,0]. During [n-T.sub.0,n-T.sub.proc,0], the TX UE decodes the received sidelink control information (SCI) from other user equipment to obtain information on the occupied resources, so as to exclude these resources. The TX UE further measures a strength of Reference Signal Received Power (RSRP) over the whole frequency band. In a case that the strength of RSRP exceeds a threshold, it is determined that the corresponding frequency domain resources are occupied by other user equipment, such that these resources are excluded. The remaining resources after the resource exclusion may be used as available candidate resources, and the candidate resources set is a set of the available candidate resources. Next, the TX UE selects one or more resources from the candidate resource set by using a random resource selection mechanism for

transmission. For example, the TX UE randomly select a resource with time domain between [n+T.sub.1,n+T.sub.2] from the candidate resource set as a resource for initially transmitting the data block and/or retransmitting the data block. For example, as shown in FIG. **2**, the TX UE selects a resource represented by a rectangular box filled with diagonal lines as the selected resource.
(23) In FIG. **2**, T.sub.0 is a threshold of a maximum range of the awareness window, T.sub.proc,0 represents a processing time at which UE decodes SCI and performs RSRP measurement, T**1** represents a processing time for the UE from the resource selection trigger n to the earliest candidate resource, T**2** represents a threshold of a maximum range of the resource selection window, which is required to be smaller than the allowable delay of the data block to be transmitted.
(24) The predetermined candidate time-frequency resource blocks mentioned above may be the remaining resources after the resource exclusion mentioned in the description with reference to FIG. **2**, or may be time-frequency resource blocks predetermined according to other methods.
(25) Hereinafter, the description of selecting a time-frequency resource block for initially transmitting a data block and/or a time-frequency resource block for retransmitting a data block from a candidate resource set is made in combination with the scenario of sidelink resource selection mode 2. However, those skilled in the art should understand that the following description of selecting a time-frequency resource block for initially transmitting a data block and/or a time-frequency resource block for retransmitting a data block is not limited to the scenario of sidelink resource selection mode 2, and may be applied to other scenarios in 4G or 5G or other communication modes where user equipment selects a time-frequency resource block for initially transmitting a data block and/or a time-frequency resource block for retransmitting a data block.
(26) As an example, the processing unit **101** may be configured to: divide a resource selection window including the candidate resource set into a predetermined number of sub-resource selection windows in a time dimension, such that each of the sub-resource selection windows has substantially the same time duration; and select a candidate time-frequency resource block, from at least one chronologically preceding sub-resource selection window which includes at least one candidate time-frequency resource block, as the time-frequency resource block for initially transmitting the data block.
(27) As an example, in the scenario of sidelink resource selection mode 2, the predetermined number may be the number of time-frequency resource blocks indicated by the sidelink control information (SCI), that is, the number of time-frequency resource blocks that may be reserved indicated by SCI. For example, the predetermined number may be a sum of the number of times for initially transmitting the data block and the number of times for retransmitting the data block, that is, the number of times for retransmitting the data block plus 1.
(28) FIG. **3** is a diagram showing an example in which a resource selection window is divided into a predetermined number of sub-resource selection windows in a time dimension according to an embodiment of the present disclosure. In FIG. **3** and other figures to be described below, unavailable time-frequency resource blocks are represented by rectangular blocks filled with gray, and available candidate time-frequency resource blocks are represented by rectangular blocks filled with white. FIG. **3** shows 9 candidate time-frequency resource blocks, that is, the candidate resource set includes 9 candidate time-frequency resource blocks. A window including all the time-frequency resource blocks shown in FIG. **3** is a resource selection window. It is assumed that the preconfigured SCI indicates 3 time-frequency resource blocks, that is, the predetermined number Nmax is equal to 3. Therefore, the resource selection window shown in FIG. **3** is divided into 3 sub-resource selection windows (a sub-resource selection window **1**, a sub-resource selection window **2** and a sub-resource selection window **3**) in a time dimension, such that each of the sub-resource selection windows has substantially the same time duration (in the example shown in FIG. **3**, each of the sub-resource selection windows has the time duration of 5 time-frequency resource

blocks). In the example shown in FIG. **3**, the sub-resource selection window **1** includes 3 candidate time-frequency resource blocks, the sub-resource selection window **2** includes 2 candidate time-frequency resource blocks, and the sub-resource selection window **3** includes 4 candidate time-frequency resource blocks.

(29) The processing unit **101** may select a candidate time-frequency resource block from the sub-resource selection window **1** as the time-frequency resource block for initially transmitting a data block (for example, the time-frequency resource block shown with a dotted box in the sub-resource selection window **1** shown in FIG. **3**). In this way, it can be ensured that the initial transmission of the data block occurs at a first 1/Nmax(⅓) of the resource selection window, such that the delay of the initial transmission of the data block can be reduced. In addition, for example, the processing unit **101** may select a candidate time-frequency resource block from the sub-resource selection window **2** as the time-frequency resource block for initially transmitting the data block. In this way, the delay of the initial transmission of the data block can be reduced, compared with a case in which a candidate time-frequency resource block is selected from the sub-resource selection window **3** as the time-frequency resource block for initially transmitting the data block. Therefore, the electronic apparatus **100** according to the embodiment of the present disclosure can reduce the delay of the initial transmission of the data block, and thus is suitable for some services with high delay requirements, such as safety information notification, in-vehicle games and the like.

(30) Those skilled in the art can understand that FIG. **3** is only an example. In other examples, the number of candidate time-frequency resource blocks included in the candidate resource set, the predetermined number Nmax, and positions of candidate time-frequency resource blocks in the resource selection window may be different from the example shown in FIG. **3**.

(31) According to the communication mode in the conventional technology (for example, sidelink resource selection mode 2), a candidate time-frequency resource block is randomly selected from the candidate resource set as the time-frequency resource block for initially transmitting a data block. However, the random selection may set the selected time-frequency resource block to be in the chronologically later part of the resource selection window, resulting in an increase in the delay of the initial transmission of the data block.

(32) However, the electronic apparatus **100** according to the embodiment of the present disclosure divides the resource selection window in the time dimension and selects a time-frequency resource block for initially transmitting the data block from the chronologically preceding sub-resource selection window, such that the delay of the initial transmission of the data block can be reduced.

(33) As an example, the processing unit **101** may be further configured to select a candidate time-frequency resource block satisfying a predetermined condition, from candidate time-frequency resource blocks chronologically after the time-frequency resource block for initially transmitting the data block in the resource selection window, as the time-frequency resource block for retransmitting the data block. Preferably, the processing unit **101** may be further configured to select a candidate time-frequency resource block satisfying a predetermined condition, from candidate time-frequency resource blocks chronologically immediately after the time-frequency resource block for initially transmitting the data block in the resource selection window, as the time-frequency resource block for retransmitting the data block.

(34) For example, in a scenario of 5G NR sidelink communication, there are two resource retransmission modes, including a blind retransmission and a retransmission based on HARQ-ACK. In the blind retransmission, the same data block (for example, a transport block (TB)) may be repeatedly transmitted using reserved resources in a case that any feedback information is not received. In the retransmission based on HARQ-ACK feedback information, the transmitter UE retransmits the data block by using the reserved resources after receiving NACK (negative acknowledgement) information. As the transmitter UE needs to wait for the feedback of ACK/NACK information, a time interval between two resource transmissions should satisfy a condition of Z=a+b, where a represents a time interval between the end of transmission of the last

symbol in initial PSSCH (Physical SideLink Shared Channel) and the start of transmission of the first symbol in PSFCH (Physical SideLink Feedback Channel) that feedbacks PSSCH, and b represents a sum of a reception time of PSFCH and a preparation time for retransmitting the data block.
(35) For example, in a case of retransmitting the data block based on a hybrid automatic repeat request acknowledgement (HARQ-ACK), satisfying the predetermined condition may refer to a case in which candidate time-frequency resource blocks satisfying a condition that the time interval between two data transmissions at least satisfies Z=a+b are used as time-frequency resource blocks for retransmitting the data block.
(36) For example, in a case of retransmitting a data block based on the blind retransmission, satisfying the predetermined condition may refer to a case in which candidate time-frequency resource blocks that may be successfully reserved in the resource selection window are used as time-frequency resource blocks for retransmitting the data block.
(37) The time-frequency resource block for retransmitting the data block may be included in the same sub-resource selection window as the time-frequency resource block for initially transmitting a data block, or may be included in a sub-resource selection window different from the sub-resource selection window including the time-frequency resource block for initially transmitting the data block.
(38) For example, in a case of Nmax being equal to 3, it is assumed that the time-frequency resource block for initially transmitting a data block is selected in the sub-resource selection window **1**. It is assumed that the sub-resource selection window **1** includes 2 candidate time-frequency resource blocks satisfying the above predetermined conditions, these two candidate time-frequency resource blocks are used as time-frequency resource blocks for retransmitting the data block. In this case, the time-frequency resource block for initially transmitting a data block and the time-frequency resource block for retransmitting the data block 2 times are both included in the sub-resource selection window **1**, such that the delay of the retransmission of the data block can be reduced. In addition, for example, even if the time-frequency resource blocks for retransmitting the data block are not included in the sub-resource selection window **1** but included in the sub-resource selection window **2**, the delay of the retransmission of the data block can be reduced compared with a case in which the time-frequency resource blocks for retransmitting the data block is randomly selected from the sub-resource selection window **3**.
(39) As an example, the processing unit **101** may be configured to: divide a resource selection window including the candidate resource set into a predetermined number of sub-resource selection windows, such that each of the sub-resource selection windows includes substantially the same number of candidate time-frequency resource blocks; and select a candidate time-frequency resource block from a most chronologically preceding sub-resource selection window as the time-frequency resource block for initially transmitting the data block.
(40) As an example, in the scenario of sidelink resource selection mode 2, the predetermined number may be the number of time-frequency resource blocks indicated by the sidelink control information (SCI), that is, the number of time-frequency resource blocks that may be reserved indicated by SCI. For example, the predetermined number may be a sum of the number of times for initially transmitting the data block and the number of times for retransmitting the data block, that is, the number of times for retransmitting the data block plus 1.
(41) FIG. **4** is a diagram showing an example in which a resource selection window is divided based on the number of candidate time-frequency resource blocks according to an embodiment of the present disclosure. The resource selection window in FIG. **4** is the same as that in FIG. **3**, which is not be repeated here. As shown in FIG. **4**, the resource selection window is divided into 3 sub-resource selection windows (a sub-resource selection window **1′**, a sub-resource selection window **2′** and a sub-resource selection window **3′**), such that each of the sub-resource selection windows includes the same number of candidate time-frequency resource blocks (in the example shown in

FIG. **4**, each of the sub-resource selection windows includes 3 candidate time-frequency resource blocks).
(42) The processing unit **101** selects a candidate time-frequency resource block from the sub-resource selection window **1′** as a time-frequency resource block for initially transmitting the data block (for example, the time-frequency resource block shown with a dotted box in the sub-resource selection window **1′** shown in FIG. **4**). In this way, the delay of the initial transmission of the data block can be reduced, such that the electronic apparatus **100** according to the embodiment of the present disclosure is suitable for some services with high delay requirements, such as safety information notification, in-vehicle games and the like.
(43) As an example, the processing unit **101** may be further configured to select a candidate time-frequency resource block satisfying a predetermined condition, from candidate time-frequency resource blocks chronologically after the time-frequency resource block for initially transmitting the data block in the resource selection window, as the time-frequency resource block for retransmitting the data block. Preferably, the processing unit **101** may be further configured to select a candidate time-frequency resource block satisfying a predetermined condition, from candidate time-frequency resource blocks chronologically immediately after the time-frequency resource block for initially transmitting the data block in the resource selection window, as the time-frequency resource block for retransmitting the data block. The predetermined condition is the same as the predetermined condition described above, which are not repeated here. The time-frequency resource block for retransmitting the data block may be included in the same sub-resource selection window as the time-frequency resource block for initially transmitting the data block, or may be included in a sub-resource selection window different from the sub-resource selection window including the time-frequency resource block for initially transmitting the data block. For example, in a case of Nmax being equal to 3, it is assumed that the sub-resource selection window **1′** includes 2 candidate time-frequency resource blocks satisfying the above predetermined conditions, these two candidate time-frequency resource blocks are used as a time-frequency resource block for retransmitting the data block. In this case, the time-frequency resource block for initially transmitting the data block and the time-frequency resource block for retransmitting the data block 2 times are both included in the sub-resource selection window **1′**, such that the delay of the retransmission of the data block can be reduced. In addition, for example, even if the time-frequency resource blocks for retransmitting the data block are not included in the sub-resource selection window **1′** but included in the sub-resource selection window **2′**, the delay of retransmission of the data blocks can be reduced compared with a case in which the time-frequency resource blocks for retransmitting the data block is randomly selected from the sub-resource selection window **3′**.
(44) As an example, in a case of Nmax being equal to 3, each electronic apparatus selects the time-frequency resource block for initially transmitting the data block from the sub-resource selection window **1′**, and selects the time-frequency resource block for retransmitting the data block from the sub-resource selection window **2′** and sub-resource selection window **3′** respectively, such that the collision in a case that different electronic apparatuses select resources can be avoided.
(45) As an example, in addition to the predetermined candidate time-frequency resource blocks, the candidate resource set further includes one or more candidate time-frequency resource blocks included in an abnormal resource pool which is different from a preconfigured resource pool including the predetermined candidate time-frequency resource blocks, and the resource selection window further includes the one or more candidate time-frequency resource blocks included in the abnormal resource pool.
(46) FIG. **5** is a diagram showing an example in which a resource selection window further includes one or more candidate time-frequency resource blocks included in an abnormal resource pool according to an embodiment of the present disclosure. 9 time-frequency resource blocks filled with white shown in FIG. **5** correspond to 9 candidate time-frequency resource blocks shown in

FIG. **3** and FIG. **4**, and the normal resource pool shown in FIG. **5** corresponds to the resource selection window shown in FIG. **3** and FIG. **4**. 3 time-frequency resource blocks filled with black shown in FIG. **5** are available candidate time-frequency resource blocks included in the abnormal resource pool, and the time-frequency resource blocks filled with gray shown in FIG. **5** are unavailable time-frequency resource blocks. Compared with the examples described with reference to FIG. **3** and FIG. **4**, in FIG. **5**, the candidate resource set further includes 3 candidate time-frequency resource blocks included in the abnormal resource pool which is different from the normal resource pool, and the resource selection window further includes 3 candidate time-frequency resource blocks included in the abnormal resource pool.

(47) The abnormal resource pool may enrich the composition of the candidate resource set, and may assist in the selection of the time-frequency resource block for initially transmitting the data block and/or the time-frequency resource block for retransmitting the data block, that is, the abnormal resource pool may increase the selectivity of the time-frequency resource block for initially transmitting the data block and/or the time-frequency resource block for retransmitting the data block. In addition, the abnormal resource pool may be customized based on the characteristics of different data services. In addition, since the candidate time-frequency resource blocks included in the abnormal resource pool may chronologically precede the candidate time-frequency resource blocks included in the normal resource pool, the delay of the initial transmission of the data block and/or the retransmission of the data block can be further reduced. For example, in the example of FIG. **5**, the resource selection window is divided into sub-resource selection window **1**, sub-resource selection window **2** and sub-resource selection window **3** in a time dimension, such that each of the sub-resource selection windows has substantially the same time duration (in the example shown in FIG. **5**, each of the sub-resource selection windows has the time duration of 5 time-frequency resource blocks). The sub-resource selection window **1** includes 4 candidate time-frequency resource blocks, the sub-resource selection window **2** includes 3 candidate time-frequency resource blocks, and the sub-resource selection window **3** includes 5 candidate time-frequency resource blocks. Since the candidate time-frequency resource block included in the abnormal resource pool shown with dotted box in FIG. **5** chronologically precedes the candidate time-frequency resource blocks included in the normal resource pool, selecting this candidate time-frequency resource as the time-frequency resource block for initially transmitting the data block can further reduce the delay of the initial transmission of the data block. For the selection of time-frequency resource blocks for retransmitting the data block in the example of FIG. **5**, except that the resource selection window further includes one or more candidate time-frequency resource blocks included in the abnormal resource pool, for other descriptions, reference may be made to the description of the example in FIG. **3**, which are not repeated here.

(48) FIG. **6** is a diagram showing another example in which a resource selection window further includes one or more candidate time-frequency resource blocks included in an abnormal resource pool according to an embodiment of the present disclosure. The normal resource pool and the abnormal resource pool shown in FIG. **6** correspond to the normal resource pool and the abnormal resource pool shown in FIG. **5**. In the example shown in FIG. **6**, the resource selection window including one or more candidate time-frequency resource blocks in the abnormal resource pool is divided into 3 sub-resource selection windows (a sub-resource selection window **1**′, a sub-resource selection window **2**′ and a sub-resource selection window **3**′) based on the number of candidate time-frequency resource blocks, such that each of the sub-resource selection windows includes the same number of candidate time-frequency resource blocks (in the example shown in FIG. **6**, each of the sub-resource selection windows includes 4 candidate time-frequency resource blocks).

(49) The processing unit **101** may select a candidate time-frequency resource block from the sub-resource selection window **1**′ as a time-frequency resource block for initially transmitting the data block (for example, a time-frequency resource block shown with a dotted box in the sub-resource selection window **1**′ shown in FIG. **6**), such that the delay of the initial transmission of the data

block can be further reduced.

(50) For the selection of the time-frequency resource block for retransmitting the data block in the example in FIG. **6**, except that the resource selection window further includes one or more candidate time-frequency resource blocks included in the abnormal resource pool, for other descriptions, reference may be made to the description of the example in FIG. **4**, which are not repeated here.

(51) As mentioned above, in a case of retransmitting the data block based on HARQ-ACK, the time interval between two data transmissions should satisfy a condition of Z=a+b, that is, it is required to select the time-frequency resource blocks satisfying the time interval Z from the candidate resource set as the time-frequency resource blocks for retransmitting the data block. However, in this case, some services with strict delay requirements cannot be satisfied.

(52) The processing unit **101** may be configured to: between retransmitting the data block respectively using two time-frequency resource blocks based on HARQ-ACK, perform at least one blind retransmission of the data block, so as to reduce the delay of the transmission, thereby satisfying some services with strict delay requirements. In addition, in a hybrid retransmission mechanism using a combination of the blind retransmission and the retransmission based on HARQ-ACK, since it is not required for the blind retransmission to satisfy the requirements of time interval Z, the number of times of the transmission of the data block in the same time period is increased, thereby improving the reliability of the transmission of the data block.

(53) As an example, the processing unit **101** may be configured to: between retransmitting the data block respectively using two time-frequency resource blocks based on HARQ-ACK, perform at least one blind retransmission of the data block using at least one candidate time-frequency resource block included in the abnormal resource pool which is different from the preconfigured resource pool including the predetermined candidate time-frequency resource blocks.

(54) As described above, since at least one blind retransmission of the data block is performed between retransmitting the data block respectively using two time-frequency resource blocks based on HARQ-ACK, the delay of the retransmission of the data block can be reduced and the reliability of the transmission of the data block is increased.

(55) In addition, since the blind retransmission may be performed by using the candidate time-frequency resource blocks included in the abnormal resource pool, the selectivity of time-frequency resource blocks for the blind retransmission of data blocks is increased.

(56) As mentioned above, in a case of retransmitting the data block based on HARQ-ACK, the time interval between two data transmissions should satisfy a condition of Z=a+b, that is, it is required to select the time-frequency resource blocks satisfying the time interval Z from the candidate resource set as the time-frequency resource blocks for retransmitting the data block. However, in some cases, the time-frequency resources satisfying the time interval Z may not be found in the candidate resource set (for example, the time interval Z cannot be satisfied between the time-frequency resource block for initially transmitting the data block and the time-frequency resource block for retransmitting the data block, and/or the time interval Z cannot be satisfied between the time-frequency resource blocks for retransmitting the data block). In the conventional technology, if the time-frequency resources satisfying the time interval Z cannot be found in the candidate resource set, the problem can be autonomously solved by the behavior of the transmitter UE, but the time-frequency resource block for retransmitting the data block have to satisfy the requirement of the time interval Z. However, the specific behavior of the transmitter UE is not confirmed in the conventional technology. Hereinafter, the specific behavior of the electronic apparatus **100** according to the embodiment of the present disclosure in a case that the time-frequency resources satisfying the time interval Z cannot be found in the candidate resource set will be described.

(57) As an example, the processing unit **101** may be configured to: in a case that the candidate resource set does not include candidate time-frequency resource blocks which satisfy a predetermined retransmission time interval for retransmitting the data block based on HARQ-ACK,

increase a threshold of a reference signal received power (RSRP) to increase the number of the candidate time-frequency resource blocks in the candidate resource set to select candidate time-frequency resource blocks satisfying the predetermined retransmission time interval from the candidate resource set including the increased number of candidate time-frequency resource blocks for retransmitting the data block. The processing circuit is configured to include a time-frequency resource block having a RSRP lower than the threshold as a candidate time-frequency resource block in the candidate resource set in a resource awareness period in which the candidate time-frequency resource block is predetermined.

(58) For example, the predetermined retransmission time interval is the time interval Z mentioned above.

(59) In the resource awareness period described with reference to FIG. **2** for example, the electronic apparatus **100** is configured to include a time-frequency resource block having a RSRP lower than a threshold as a candidate time-frequency resource block in the candidate resource set and exclude a time-frequency resource block having a RSRP higher than or equal to the threshold from the candidate resource set by comparing the RSRP measurement values of the candidate time-frequency resource blocks with the threshold, so as to screen out the candidate time-frequency resource blocks. Therefore, by increasing the threshold of RSRP, the number of candidate time-frequency resource blocks included in the candidate resource set can be increased, such that candidate time-frequency resource blocks satisfying the predetermined retransmission time interval are selected from the candidate resource set including the increased number of candidate time-frequency resource blocks.

(60) As an example, the processing unit **101** may be configured to: in a case that the candidate resource set does not include candidate time-frequency resource blocks which satisfy a predetermined retransmission time interval for retransmitting the data block based on HARQ-ACK, select candidate time-frequency resource blocks satisfying the predetermined retransmission time interval from at least one candidate time-frequency resource block included in the abnormal resource pool which is different from the preconfigured resource pool including the predetermined candidate time-frequency resource blocks for retransmitting the data block.

(61) At least one candidate time-frequency resource block included in the abnormal resource pool increases the number of available candidate time-frequency resource blocks. Therefore, it may be possible to select a candidate time-frequency resource block satisfying the predetermined retransmission time interval from the abnormal resource pool.

(62) As an example, the processing unit **101** may be configured to: in a case that the candidate resource set does not include candidate time-frequency resource blocks satisfying a predetermined retransmission time interval for retransmitting the data block based on HARQ-ACK, change a method for retransmitting the data block by adding a field that indicates a method for retransmitting a data block in control information of data transmission. The field is used to indicate that the retransmission of the data block is performed based on one of HARQ-ACK, a blind retransmission as well as a hybrid retransmission including HARQ-ACK and the blind retransmission.

(63) For example, in a case that the candidate resource set does not include candidate time-frequency resource blocks satisfying a predetermined retransmission time interval for retransmitting the data block based on HARQ-ACK, the processing unit **101** may indicate the change in the current retransmission method by the above-mentioned field that indicates the method for retransmitting the data block (for example, two bits may be used to indicate that the current retransmission is performed based on HARQ-ACK, the blind retransmission or the hybrid retransmission). In a case that the candidate resource set does not include candidate time-frequency resource blocks satisfying a predetermined retransmission time interval for retransmitting the data block based on HARQ-ACK, the normal data transmission may be guaranteed by changing the current retransmission method (for example, change to the blind retransmission or the hybrid retransmission), thereby reducing the delay of the transmission of the data block.

(64) As an example, the above control information is SCI. For example, the method for retransmitting the data block is changed by adding a field that indicates the method for retransmitting the data block in SCI.
(65) Those skilled in the art may understand that the above-described embodiments of selecting a time-frequency resource block for initially transmitting a data block, selecting a time-frequency resource block for retransmitting a data block, and selecting a time-frequency resource block for retransmitting a data block in a case that the candidate resource set does not include candidate time-frequency resource blocks satisfying a predetermined retransmission time interval for retransmitting the data block based on HARQ-ACK may be embodied independently or combined with each other.
(66) In the process of describing the electronic apparatus for wireless communications in the above embodiments, some processes or methods are also disclosed. Hereinafter, an overview of the methods is given without repeating some details that are discussed above. However, it should be noted that, although the methods are disclosed while describing the electronic apparatus for wireless communications, the methods do not necessarily employ or are not necessarily performed by the aforementioned components. For example, the embodiments of the electronic apparatus for wireless communications may be partially or completely implemented with hardware and/or firmware, and the method for wireless communications described below may be performed by a computer-executable program completely, although the hardware and/or firmware for the electronic apparatus for wireless communications may also be used in the methods.
(67) FIG. **7** shows a flowchart of a method **700** for wireless communications according to an embodiment of the present disclosure. The method **700** starts at step S**702**. In step S**704**, a time-frequency resource block for initially transmitting a data block and/or a time-frequency resource block for retransmitting a data block is selected from at least a candidate resource set comprising predetermined candidate time-frequency resource blocks. The method **700** ends at step S**706**.
(68) The method may be performed by the electronic apparatus **100** described above, for example. The specific details may be found in the description in the corresponding position above, which are not repeated here.
(69) The technology according to the present disclosure may be applied to various products.
(70) The electronic apparatus **100** may be implemented as various user equipment. The user equipment may be implemented as a mobile terminal (such as a smart phone, a tablet personal computer (PC), a notebook PC, a portable game terminal, a portable/dongle type mobile router, and a digital camera) or a in-vehicle terminal (such as an vehicle navigation device). The user equipment may also be implemented as a terminal that executes Machine-to-Machine (M2M) communications (which is also referred to as a machine type communication (MTC) terminal). In addition, the user equipment may be a wireless communication module (such as an integrated circuit module including a single chip) installed on each of the above-mentioned terminals.
Application Examples of Base Station
(71) (First Application Example)
(72) FIG. **8** is a block diagram showing a first example of a schematic configuration of an eNB or gNB to which the technology according to the present disclosure may be applied. Note that, the following description takes an eNB as an example, which may also be applied to a gNB. An eNB **800** includes one or more antennas **810** and a base station device **820**. The base station device **820** and each antenna **810** may be connected to each other via an RF cable.
(73) Each of the antennas **810** includes a single antenna element or multiple antenna elements (such as multiple antenna elements included in a Multi-Input Multi-Output (MIMO) antenna), and is used for the base station device **820** to transmit and receive wireless signals. As shown in FIG. **8**, the eNB **800** may include multiple antennas **810**. For example, the multiple antennas **810** may be compatible with multiple frequency bands used by the eNB **800**. Although FIG. **8** shows an example in which the eNB **800** includes multiple antennas **810**, the eNB **800** may also include a

single antenna **810**.

(74) The base station device **820** includes a controller **821**, a memory **822**, a network interface **823**, and a radio communication interface **825**.

(75) The controller **821** may be, for example, a CPU or a DSP, and manipulate various functions of a higher layer of the base station device **820**. For example, the controller **821** generates a data packet based on data in a signal processed by the radio communication interface **825**, and transfers the generated packet via the network interface **823**. The controller **821** may bundle data from multiple baseband processors to generate a bundled packet, and transfer the generated bundled packet. The controller **821** may have a logical function for performing control such as radio resource control, radio bearer control, mobility management, admission control, and scheduling. The control may be executed in conjunction with nearby eNBs or a core network node. The memory **822** includes an RAM and an ROM, and stores programs executed by the controller **821** and various types of control data (such as a terminal list, transmission power data, and scheduling data).

(76) The network interface **823** is a communication interface for connecting the base station device **820** to a core network **824**. The controller **821** may communicate with the core network node or another eNB via the network interface **823**. In this case, the eNB **800** and the core network node or other eNBs may be connected to each other via a logical interface (such as an S1 interface and an X2 interface). The network interface **823** may also be a wired communication interface, or a wireless communication interface for a wireless backhaul line. If the network interface **823** is a wireless communication interface, the network interface **823** may use a higher frequency band for wireless communications than the frequency band used by the radio communication interface **825**.

(77) The radio communication interface **825** supports any cellular communication scheme (such as Long Term Evolution (LTE) and LTE-Advanced), and provides wireless connection to a terminal located in a cell of the eNB **800** via an antenna **810**. The radio communication interface **825** may generally include, for example, a baseband (BB) processor **826** and an RF circuit **827**. The BB processor **826** may execute, for example, encoding/decoding, modulation/demodulation, and multiplexing/de-multiplexing, and execute various types of signal processing of layers (e.g., L1, Medium Access Control (MAC), Radio Link Control (RLC), and Packet Data Convergence Protocol (PDCP)). Instead of the controller **821**, the BB processor **826** may have a part or all of the above-mentioned logical functions. The BB processor **826** may be a memory storing a communication control program, or a module including a processor and related circuits configured to execute the program. An update program may cause the function of the BB processor **826** to be changed. The module may be a card or blade inserted into a slot of the base station device **820**. Alternatively, the module may also be a chip mounted on a card or blade. In addition, the RF circuit **827** may include, for example, a frequency mixer, a filter, and an amplifier, and transmit and receive a wireless signal via the antenna **810**.

(78) As shown in FIG. **8**, the radio communication interface **825** may include multiple BB processors **826**. For example, the multiple BB processors **826** may be compatible with multiple frequency bands used by the eNB **800**. As shown in FIG. **8**, the radio communication interface **825** may include multiple RF circuits **827**. For example, the multiple RF circuits **827** may be compatible with multiple antenna elements. Although FIG. **8** shows an example in which the radio communication interface **825** includes multiple BB processors **826** and multiple RF circuits **827**, the radio communication interface **825** may also include a single BB processor **826** or a single RF circuit **827**.

(79) In the eNB **800** as shown in FIG. **8**, the transceiver may be implemented by a radio communication interface **825**. At least a part of the function may also be implemented by the controller **821**.

(80) (Second Application Example)

(81) FIG. **9** is a block diagram showing a second example of a schematic configuration of an eNB

or gNB to which the technology according to the present disclosure may be applied. Note that similarly, the following description takes an eNB as an example, which may also be applied to a gNB. An eNB **830** includes one or more antennas **840**, a base station device **850**, and an RRH **860**. The RRH **860** and each antenna **840** may be connected to each other via an RF cable. The base station device **850** and the RRH **860** may be connected to each other via a high-speed line such as an optical fiber cable.

(82) Each of the antennas **840** includes a single antenna element or multiple antenna elements (such as multiple antenna elements included in a MIMO antenna) and is used for the RRH **860** to transmit and receive a wireless signal. As shown in FIG. **9**, the eNB **830** may include multiple antennas **840**. For example, the multiple antennas **840** may be compatible with multiple frequency bands used by the eNB **830**. Although FIG. **9** shows an example in which the eNB **830** includes multiple antennas **840**, the eNB **830** may also include a single antenna **840**.

(83) The base station device **850** includes a controller **851**, a memory **852**, a network interface **853**, a radio communication interface **855**, and a connection interface **857**. The controller **851**, the memory **852**, and the network interface **853** are the same as the controller **821**, the memory **822**, and the network interface **823** as described with reference to FIG. **8**.

(84) The radio communication interface **855** supports any cellular communication scheme (such as LTE and LTE-Advanced), and provides wireless communications to a terminal located in a sector corresponding to the RRH **860** via the RRH **860** and the antenna **840**. The radio communication interface **855** may generally include, for example, a BB processor **856**. The BB processor **856** is the same as the BB processor **826** as described with reference to FIG. **8** except that the BB processor **856** is connected to an RF circuit **864** of the RRH **860** via the connection interface **857**. As shown in FIG. **9**, the radio communication interface **855** may include multiple BB processors **856**. For example, the multiple BB processors **856** may be compatible with multiple frequency bands used by the eNB **830**. Although FIG. **9** shows an example in which the radio communication interface **855** includes multiple BB processors **856**, the radio communication interface **855** may also include a single BB processor **856**.

(85) The connection interface **857** is an interface for connecting the base station device **850** (radio communication interface **855**) to the RRH **860**. The connection interface **857** may also be a communication module for communication in the above-mentioned high-speed line that connects the base station device **850** (radio communication interface **855**) to the RRH **860**.

(86) The RRH **860** includes a connection interface **861** and a radio communication interface **863**.

(87) The connection interface **861** is an interface for connecting the RRH **860** (radio communication interface **863**) to the base station device **850**. The connection interface **861** may also be a communication module for communication in the above-mentioned high-speed line.

(88) The radio communication interface **863** transfers and receives wireless signals via the antenna **840**. The radio communication interface **863** may generally include, for example, an RF circuit **864**. The RF circuit **864** may include, for example, a frequency mixer, a filter, and an amplifier, and transfer and receive wireless signals via the antenna **840**. As shown in FIG. **9**, the radio communication interface **863** may include multiple RF circuits **864**. For example, the multiple RF circuits **864** may support multiple antenna elements. Although FIG. **9** shows an example in which the radio communication interface **863** includes multiple RF circuits **864**, the radio communication interface **863** may also include a single RF circuit **864**.

(89) In the eNB **830** as shown in FIG. **9**, the transceiver may be implemented by the radio communication interface **855**. At least a part of the function may also be implemented by the controller **851**.

Application Example of User Equipment

(90) (First Application Example)

(91) FIG. **10** is a block diagram showing an example of a schematic configuration of a smart phone **900** to which the technology according to the present disclosure may be applied. The smart phone

900 includes a processor **901**, a memory **902**, a storage **903**, an external connection interface **904**, an camera **906**, a sensor **907**, a microphone **908**, an input device **909**, a display device **910**, a speaker **911**, a radio communication interface **912**, one or more antenna switches **915**, one or more antennas **916**, a bus **917**, a battery **918**, and an auxiliary controller **919**.
(92) The processor **901** may be, for example, a CPU or a system on a chip (SoC), and controls the functions of the application layer and other layers of the smart phone **900**. The memory **902** includes an RAM and an ROM, and stores programs executed by the processor **901** and data. The storage **903** may include a storage medium such as a semiconductor memory and a hard disk. The external connection interface **904** is an interface for connecting an external device (such as a memory card and a universal serial bus (USB) device) to the smart phone **900**.
(93) The camera **906** includes an image sensor (such as a charge coupled device (CCD) and a complementary metal oxide semiconductor (CMOS)), and generates a captured image. The sensor **907** may include a group of sensors, such as a measurement sensor, a gyro sensor, a geomagnetic sensor, and an acceleration sensor. The microphone **908** converts sound input to the smart phone **900** into an audio signal. The input device **909** includes, for example, a touch sensor configured to detect a touch on a screen of the display device **910**, a keypad, a keyboard, a button, or a switch, and receives an operation or information input from the user. The display device **910** includes a screen (such as a liquid crystal display (LCD) and an organic light emitting diode (OLED) display), and displays an output image of the smart phone **900**. The speaker **911** converts the audio signal output from the smart phone **900** into sound.
(94) The radio communication interface **912** supports any cellular communication scheme (such as LTE and LTE-Advanced), and executes wireless communications. The radio communication interface **912** may generally include, for example, a BB processor **913** and an RF circuit **914**. The BB processor **913** may execute, for example, encoding/decoding, modulation/demodulation, and multiplexing/de-multiplexing, and execute various types of signal processing for wireless communications. In addition, the RF circuit **914** may include, for example, a frequency mixer, a filter, and an amplifier, and transfer and receive wireless signals via the antenna **916**. Note that, although the figure shows a case where one RF link is connected with one antenna, this is only schematic, and a case where one RF link is connected with multiple antennas via multiple phase shifters is also included. The radio communication interface **912** may be a chip module on which the BB processor **913** and the RF circuit **914** are integrated. As shown in FIG. **10**, the radio communication interface **912** may include multiple BB processors **913** and multiple RF circuits **914**. Although FIG. **10** shows an example in which the radio communication interface **912** includes multiple BB processors **913** and multiple RF circuits **914**, the radio communication interface **912** may also include a single BB processor **913** or a single RF circuit **914**.
(95) Furthermore, in addition to the cellular communication scheme, the radio communication interface **912** may support other types of wireless communication schemes, such as a short-range wireless communication scheme, a near field communication scheme, and a wireless local area network (LAN) scheme. In this case, the radio communication interface **912** may include a BB processor **913** and an RF circuit **914** for each wireless communication scheme.
(96) Each of the antenna switches **915** switches a connection destination of the antenna **916** among multiple circuits included in the radio communication interface **912** (for example, circuits for different wireless communication schemes).
(97) Each of the antennas **916** includes a single antenna element or multiple antenna elements (such as multiple antenna elements included in a MIMO antenna), and is used for the radio communication interface **912** to transmit and receive wireless signals. As shown in FIG. **10**, the smart phone **900** may include multiple antennas **916**. Although FIG. **10** shows an example in which the smart phone **900** includes multiple antennas **916**, the smart phone **900** may also include a single antenna **916**.
(98) Furthermore, the smart phone **900** may include an antenna **916** for each wireless

communication scheme. In this case, the antenna switch **915** may be omitted from the configuration of the smart phone **900**.
(99) The bus **917** connects the processor **901**, the memory **902**, the storage **903**, the external connection interface **904**, the camera **906**, the sensor **907**, the microphone **908**, the input device **909**, the display device **910**, the speaker **911**, the radio communication interface **912**, and the auxiliary controller **919** to each other. The battery **918** supplies power to each block of the smart phone **900** as shown in FIG. **10** via a feeder line, which is partially shown as a dashed line in the figure. The auxiliary controller **919** manipulates the least necessary function of the smart phone **900** in a sleep mode, for example.
(100) In the smart phone **900** as shown in FIG. **10**, in a case that the electronic apparatus **100** described with reference to FIG. **1** is implemented as user equipment, a transceiver of the electronic apparatus **100** may be implemented by the radio communication interface **912**. At least a part of the function may also be implemented by the processor **901** or the auxiliary controller **919**. For example, the processor **901** or the auxiliary controller **919** may select a time-frequency resource block for initially transmitting a data block and/or a time-frequency resource block for retransmitting a data block by using the processing unit **101** described with reference to FIG. **1**.
(101) (Second Application Example)
(102) FIG. **11** is a block diagram showing an example of a schematic configuration of a vehicle navigation device to which the technology according to the present disclosure may be applied. The vehicle navigation device **920** includes a processor **921**, a memory **922**, a global positioning system (GPS) module **924**, a sensor **925**, a data interface **926**, a content player **927**, a storage medium interface **928**, an input device **929**, a display device **930**, a speaker **931**, a radio communication interface **933**, one or more antenna switches **936**, one or more antennas **937**, and a battery **938**.
(103) The processor **921** may be, for example, a CPU or a SoC, and controls the navigation function and additional functions of the vehicle navigation device **920**. The memory **922** includes an RAM and an ROM, and stores programs executed by the processor **921** and data.
(104) The GPS module **924** uses a GPS signal received from a GPS satellite to measure a position of the vehicle navigation device **920** (such as latitude, longitude, and altitude). The sensor **925** may include a group of sensors, such as a gyro sensor, a geomagnetic sensor, and an air pressure sensor. The data interface **926** is connected to, for example, an in-vehicle network **941** via a terminal which is not shown, and acquires data generated by a vehicle (such as vehicle speed data).
(105) The content player **927** reproduces content stored in a storage medium (such as a CD and a DVD), which is inserted into the storage medium interface **928**. The input device **929** includes, for example, a touch sensor configured to detect a touch on a screen of the display device **930**, a button, or a switch, and receives an operation or information input from the user. The display device **930** includes a screen such as an LCD or OLED display, and displays an image of a navigation function or reproduced content. The speaker **931** outputs the sound of the navigation function or the reproduced content.
(106) The radio communication interface **933** supports any cellular communication scheme (such as LTE and LTE-Advanced), and executes wireless communication. The radio communication interface **933** may generally include, for example, a BB processor **934** and an RF circuit **935**. The BB processor **934** may execute, for example, encoding/decoding, modulation/demodulation, and multiplexing/de-multiplexing, and execute various types of signal processing for wireless communications. In addition, the RF circuit **935** may include, for example, a frequency mixer, a filter, and an amplifier, and transfer and receive wireless signals via the antenna **937**. The radio communication interface **933** may also be a chip module on which the BB processor **934** and the RF circuit **935** are integrated. As shown in FIG. **11**, the radio communication interface **933** may include multiple BB processors **934** and multiple RF circuits **935**. Although FIG. **11** shows an example in which the radio communication interface **933** includes multiple BB processors **934** and multiple circuits **935**, the radio communication interface **933** may also include a single BB

processor **934** or a single RF circuit **935**.
(107) Furthermore, in addition to the cellular communication scheme, the radio communication interface **933** may support other types of wireless communication schemes, such as a short-range wireless communication scheme, a near field communication scheme, and a wireless LAN scheme. In this case, the radio communication interface **933** may include a BB processor **934** and an RF circuit **935** for each wireless communication scheme.
(108) Each of the antenna switches **936** switches a connection destination of the antenna **937** among multiple circuits included in the radio communication interface **933** (for example, circuits for different wireless communication schemes).
(109) Each of the antennas **937** includes a single antenna element or multiple antenna elements (such as multiple antenna elements included in a MIMO antenna), and is used for the radio communication interface **933** to transfer and receive wireless signals. As shown in FIG. **11**, the vehicle navigation device **920** may include multiple antennas **937**. Although FIG. **11** shows an example in which the vehicle navigation device **920** includes multiple antennas **937**, the vehicle navigation device **920** may also include a single antenna **937**.
(110) Furthermore, the vehicle navigation device **920** may include an antenna **937** for each wireless communication scheme. In this case, the antenna switch **936** may be omitted from the configuration of the vehicle navigation device **920**.
(111) The battery **938** supplies power to each block of the vehicle navigation device **920** as shown in FIG. **11** via a feeder line, which is partially shown as a dashed line in the figure. The battery **938** accumulates electric power supplied from the vehicle.
(112) In the vehicle navigation device **920** as shown in FIG. **11**, in a case that the electronic apparatus **100** described with reference to FIG. **1** is implemented as user equipment, a transceiver of the electronic apparatus **100** may be implemented by the radio communication interface **933**. At least a part of the function may also be implemented by the processor **921**. For example, the processor **921** may select a time-frequency resource block for initially transmitting a data block and/or a time-frequency resource block for retransmitting a data block by using the processing unit **101** described with reference to FIG. **1**.
(113) The technology according to the present disclosure may also be implemented as an in-vehicle system (or vehicle) **940** including one or more blocks in the vehicle navigation device **920**, the in-vehicle network **941**, and the vehicle module **942**. The vehicle module **942** generates vehicle data (such as vehicle speed, engine speed, and failure information), and outputs the generated data to the in-vehicle network **941**.
(114) The basic principle of the present disclosure has been described above in conjunction with specific embodiments. However, it should be pointed out that, for those skilled in the art, it could be understood that all or any step or component of the methods and devices according to the present disclosure may be implemented in any computing device (including processors, storage media, and the like) or network of computing devices in the form of hardware, firmware, software, or a combination thereof. This can be achieved by those skilled in the art utilizing their basic circuit design knowledge or basic programming skills after reading the description of the present disclosure.
(115) Moreover, a program product storing a machine-readable instruction code is further proposed according to the present disclosure. The instruction code, when read and executed by a machine, can execute the above-mentioned methods according to the embodiment of the present disclosure.
(116) Accordingly, a storage medium for carrying the above-mentioned program product storing a machine-readable instruction code is also included in the disclosure of the present disclosure. The storage medium includes, but is not limited to, a floppy disk, an optical disk, a magneto-optical disk, a memory card, a memory stick, and the like.
(117) In a case that the present disclosure is implemented by a software or a firmware, a program constituting the software is installed from a storage medium or a network to a computer with a

dedicated hardware structure (for example, a general-purpose computer **1200** as shown in FIG. **12**), and the computer, when installed with various programs, can execute various functions and the like.

(118) In FIG. **12**, a central processing unit (CPU) **1201** executes various processing in accordance with a program stored in a read only memory (ROM) **1202** or a program loaded from a storage part **1208** to a random access memory (RAM) **1203**. In the RAM **1203**, data required when the CPU **1201** executes various processing and the like is also stored as needed. The CPU **1201**, the ROM **1202**, and the RAM **1203** are connected to each other via a bus **1204**. The input/output interface **1205** is also connected to the bus **1204**.

(119) The following components are connected to the input/output interface **1205**: an input part **1206** (including a keyboard, a mouse, and the like), an output part **1207** (including a display, such as a cathode ray tube (CRT), a liquid crystal display (LCD), and the like, and a speaker, and the like), a storage part **1208** (including a hard disk, and the like), and a communication part **1209** (including a network interface card such as an LAN card, a modem, and the like). The communication part **1209** executes communication processing via a network such as the Internet. The driver **1210** may also be connected to the input/output interface **1205**, as needed. A removable medium **1211** such as a magnetic disk, an optical disk, a magneto-optical disk, a semiconductor memory and the like is installed on the driver **1210** as needed, such that a computer program read out therefrom is installed into the storage part **1208** as needed.

(120) In a case that the above-mentioned series of processing is implemented by a software, a program constituting the software is installed from a network such as the Internet or a storage medium such as the removable medium **1211**.

(121) Those skilled in the art should understand that, this storage medium is not limited to the removable medium **1211** as shown in FIG. **12** which has a program stored therein and which is distributed separately from an apparatus to provide a program to a user. Examples of the removable media **1211** include magnetic disks (including a floppy disk (registered trademark)), an optical disk (including a compact disk read-only memory (CD-ROM) and a digital versatile disk (DVD)), a magneto-optical disk (including a mini disk (MD) (registered trademark)), and a semiconductor memory. Alternatively, the storage medium may be the ROM **1202**, a hard disk included in the storage part **1208**, and the like, which have programs stored therein and which are distributed concurrently with the apparatus including them to users.

(122) It should also be pointed out that in the devices, methods and systems according to the present disclosure, each component or each step may be decomposed and/or recombined. These decompositions and/or recombinations should be regarded as equivalent solutions of the present disclosure. Moreover, the steps of executing the above-mentioned series of processing may naturally be executed in chronological order in the order as described, but do not necessarily need to be executed in chronological order. Some steps may be executed in parallel or independently of each other.

(123) Finally, it should be noted that, the terms “include”, “comprise” or any other variants thereof are intended to cover non-exclusive inclusion, such that a process, method, article or apparatus that includes a series of elements not only includes those elements, but also includes other elements that are not explicitly listed, or but also includes elements inherent to such a process, method, article, or apparatus. Furthermore, in the absence of more restrictions, an element defined by sentence “including one . . . ” does not exclude the existence of other identical elements in a process, method, article, or apparatus that includes the element.

(124) Although the embodiments of the present disclosure have been described above in detail in conjunction with the accompanying drawings, it should be appreciated that, the above-described embodiments are only used to illustrate the present disclosure and do not constitute a limitation to the present disclosure. For those skilled in the art, various modifications and changes may be made to the above-mentioned embodiments without departing from the essence and scope of the present

disclosure. Therefore, the scope of the present disclosure is defined only by the appended claims and equivalent meanings thereof.

(125) This technology may also be implemented as follows. (1). An electronic apparatus for wireless communications, including: a processing circuit configured to: select a time-frequency resource block for initially transmitting a data block and/or a time-frequency resource block for retransmitting a data block from at least a candidate resource set comprising predetermined candidate time-frequency resource blocks. (2). The electronic apparatus according to (1), where the processing circuit is configured to: divide a resource selection window including the candidate resource set into a predetermined number of sub-resource selection windows in a time dimension, such that each of the sub-resource selection windows has substantially the same time duration; and select a candidate time-frequency resource block, from at least one chronologically preceding sub-resource selection window which includes at least one candidate time-frequency resource block, as the time-frequency resource block for initially transmitting the data block. (3). The electronic apparatus according to (2), wherein the processing circuit is further configured to: select a candidate time-frequency resource block satisfying a predetermined condition, from candidate time-frequency resource blocks chronologically after the time-frequency resource block for initially transmitting the data block in the resource selection window, as the time-frequency resource block for retransmitting the data block. (4). The electronic apparatus according to (1), where the processing circuit is configured to: divide a resource selection window including the candidate resource set into a predetermined number of sub-resource selection windows, such that each of the sub-resource selection windows includes substantially the same number of candidate time-frequency resource blocks; and select a candidate time-frequency resource block from a most chronologically preceding sub-resource selection window as the time-frequency resource block for initially transmitting the data block. (5). The electronic apparatus according to (4), where the processing circuit is further configured to: select a candidate time-frequency resource block satisfying a predetermined condition, from candidate time-frequency resource blocks chronologically after the time-frequency resource block for initially transmitting the data block in the resource selection window, as the time-frequency resource block for retransmitting the data block. (6). The electronic apparatus according to any one of (2) to (5), where, in addition to the predetermined candidate time-frequency resource blocks, the candidate resource set further includes one or more candidate time-frequency resource blocks included in an abnormal resource pool which is different from a preconfigured resource pool including the predetermined candidate time-frequency resource blocks, and the resource selection window further includes the one or more candidate time-frequency resource blocks in the abnormal resource pool. (7). The electronic apparatus according to any one of (1) to (6), where the processing circuit is configured to: between retransmitting the data block respectively using two time-frequency resource blocks based on a hybrid automatic repeat request acknowledgement (HARQ-ACK), perform at least one blind retransmission of the data block using at least one candidate time-frequency resource block included in the abnormal resource pool which is different from the preconfigured resource pool including the predetermined candidate time-frequency resource blocks. (8). The electronic apparatus according to any one of (1) to (7), where, the processing circuit is configured to: in a case that the candidate resource set does not include candidate time-frequency resource blocks which satisfy a predetermined retransmission time interval for retransmitting the data block based on a hybrid automatic repeat request acknowledgement (HARQ-ACK), increase a threshold of a reference signal received power (RSRP) to increase the number of the candidate time-frequency resource blocks in the candidate resource set to select candidate time-frequency resource blocks satisfying the predetermined retransmission time interval from the candidate resource set including the increased number of candidate time-frequency resource blocks for retransmitting the data block, where the processing circuit is configured to include a time-frequency resource block having a RSRP lower than the threshold as a candidate time-frequency resource block in the candidate

resource set in a resource awareness period in which the candidate time-frequency resource block is predetermined. (9). The electronic apparatus according to any one of (1) to (7), where the processing circuit is configured to: in a case that the candidate resource set does not include candidate time-frequency resource blocks which satisfy a predetermined retransmission time interval for retransmitting the data block based on a hybrid automatic repeat request acknowledgement (HARQ-ACK), select candidate time-frequency resource blocks satisfying the predetermined retransmission time interval from at least one candidate time-frequency resource block included in the abnormal resource pool which is different from the preconfigured resource pool including the predetermined candidate time-frequency resource blocks for retransmitting the data block. (10). The electronic apparatus according to any one of (1) to (7), where, the processing circuit is configured to: in a case that the candidate resource set does not include candidate time-frequency resource blocks satisfying a predetermined retransmission time interval for retransmitting the data block based on a hybrid automatic repeat request acknowledgement (HARQ-ACK), change a method for retransmitting the data block by adding a field that indicates a method for retransmitting the data block in control information of data transmission, where the field is used to indicate that the retransmission of the data block is performed based on one of a hybrid automatic repeat request acknowledgement (HARQ-ACK), a blind retransmission as well as a hybrid retransmission including HARQ-ACK and blind retransmission. (11). The electronic apparatus according to (10), where the control information is sidelink control information SCI. (12). The electronic apparatus according to any one of (1) to (11), where the processing circuit is configured to select a time-frequency resource block for initially transmitting the data block and/or a time-frequency resource block for retransmitting the data block in a scenario of sidelink resource selection mode 2. (13). A method for wireless communications, including: selecting a time-frequency resource block for initially transmitting a data block and/or a time-frequency resource block for retransmitting a data block from at least a candidate resource set comprising predetermined candidate time-frequency resource blocks. (14). A computer-readable storage medium storing computer-executable instructions, where when the computer-executable instructions are executed, the method for wireless communications according to (13) is performed.

Claims

1. An electronic apparatus for wireless communications, comprising: processing circuitry configured to select a time-frequency resource block for initially transmitting a data block or a time-frequency resource block for retransmitting a data block from at least a candidate resource set including predetermined candidate time-frequency resource blocks; divide a resource selection window that includes the candidate resource set into a predetermined number of sub-resource selection windows along a time axis, such that each of the sub-resource selection windows has substantially the same time duration; and select a candidate time-frequency resource block, from at least an earliest one of the sub-resource selection windows which includes at least one candidate time-frequency resource block, as the time-frequency resource block for initially transmitting the data block, wherein the processing circuitry is further configured to in a case that the candidate resource set does not include candidate time-frequency resource blocks which satisfy a predetermined retransmission time interval for retransmitting the data block based on a hybrid automatic repeat request acknowledgement (HARQ-ACK), increase a threshold of a reference signal received power (RSRP) to increase the number of the candidate time-frequency resource blocks in the candidate resource set, and select candidate time-frequency resource blocks satisfying the predetermined retransmission time interval from the candidate resource set that includes an increased number of candidate time-frequency resource blocks for retransmitting the data block, and select a time-frequency resource block that has a RSRP lower than the threshold as a candidate time-frequency resource block in the candidate resource set in a resource awareness period in

which the candidate time-frequency resource block is predetermined.

2. The electronic apparatus according to claim 1, wherein the processing circuitry is further configured to select a candidate time-frequency resource block satisfying a predetermined condition, from candidate time-frequency resource blocks that follow the time-frequency resource block that is selected for initially transmitting the data block in the resource selection window, as the time-frequency resource block for retransmitting the data block.

3. The electronic apparatus according to claim 1, wherein in addition to the predetermined candidate time-frequency resource blocks, the candidate resource set further includes one or more candidate time-frequency resource blocks included in an abnormal resource pool which is different from a preconfigured resource pool that includes the predetermined candidate time-frequency resource blocks, and the resource selection window further includes the one or more candidate time-frequency resource blocks included in the abnormal resource pool.

4. The electronic apparatus according to claim 1, wherein the processing circuitry is further configured to between retransmitting the data block twice respectively using two time-frequency resource blocks based on a hybrid automatic repeat request acknowledgement (HARQ-ACK), perform at least one blind retransmission of the data block using at least one candidate time-frequency resource block included in an abnormal resource pool which is different from a preconfigured resource pool that includes the predetermined candidate time-frequency resource blocks.

5. An electronic apparatus for wireless communications, comprising: processing circuitry configured to select a time-frequency resource block for initially transmitting a data block or a time-frequency resource block for retransmitting a data block from at least a candidate resource set including predetermined candidate time-frequency resource blocks; divide a resource selection window that includes the candidate resource set into a predetermined number of sub-resource selection windows, such that each of the sub-resource selection windows includes substantially the same number of candidate time-frequency resource blocks; and select a candidate time-frequency resource block from an earliest one of the sub-resource selection windows as the time-frequency resource block for initially transmitting the data block, wherein the processing circuitry is further configured to in a case that the candidate resource set does not include candidate time-frequency resource blocks which satisfy a predetermined retransmission time interval for retransmitting the data block based on a hybrid automatic repeat request acknowledgement (HARQ-ACK), select candidate time-frequency resource blocks satisfying the predetermined retransmission time interval from at least one candidate time-frequency resource block included in an abnormal resource pool which is different from a preconfigured resource pool that includes the predetermined candidate time-frequency resource blocks for retransmitting the data block.

6. The electronic apparatus according to claim 5, wherein the processing circuitry is further configured to select a candidate time-frequency resource block satisfying a predetermined condition, from candidate time-frequency resource blocks that follow the time-frequency resource block that is selected for initially transmitting the data block in the resource selection window, as the time-frequency resource block for retransmitting the data block.

7. The electronic apparatus according to claim 5, wherein the processing circuitry is further configured to in a case that the candidate resource set does not include candidate time-frequency resource blocks satisfying a predetermined retransmission time interval for retransmitting the data block based on a hybrid automatic repeat request acknowledgement (HARQ-ACK), change a method for retransmitting the data block by adding a field that indicates a method for retransmitting the data block in control information of data transmission, and the field is used to indicate that the retransmission of the data block is performed based on one of a hybrid automatic repeat request acknowledgement (HARQ-ACK), a blind retransmission, and a hybrid retransmission including the HARQ-ACK and the blind retransmission.

8. The electronic apparatus according to claim 7, wherein the control information is sidelink control

information SCI.

9. The electronic apparatus according to claim 5, wherein the processing circuitry is further configured to select a time-frequency resource block for initially transmitting the data block or a time-frequency resource block for retransmitting the data block in a scenario of sidelink resource selection mode 2.